Kickstarter

DATA 549 Final Project

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Kickstarter

The company was launched in 2009

Goal of the company is to provide backing on projects that can come from artists , scientists, or entrepreneurs.

Donors will either back or not back the project.

Why?

Crowdfunding allows for brilliant ideas to come to life. These projects are either products of startups, social issue groups, or entrepreneurs.

Can be for profit or non-profit

This classification project will focus to see if we can predict if backers will say yes or no.

Data Description

It is a Kaggle public dataset of project campaigns from year 2017. Contains 215,513 rows each containing a unique description of the project and a binary state of the funding campaign i.e., successful or failed.

Data has no missing values.

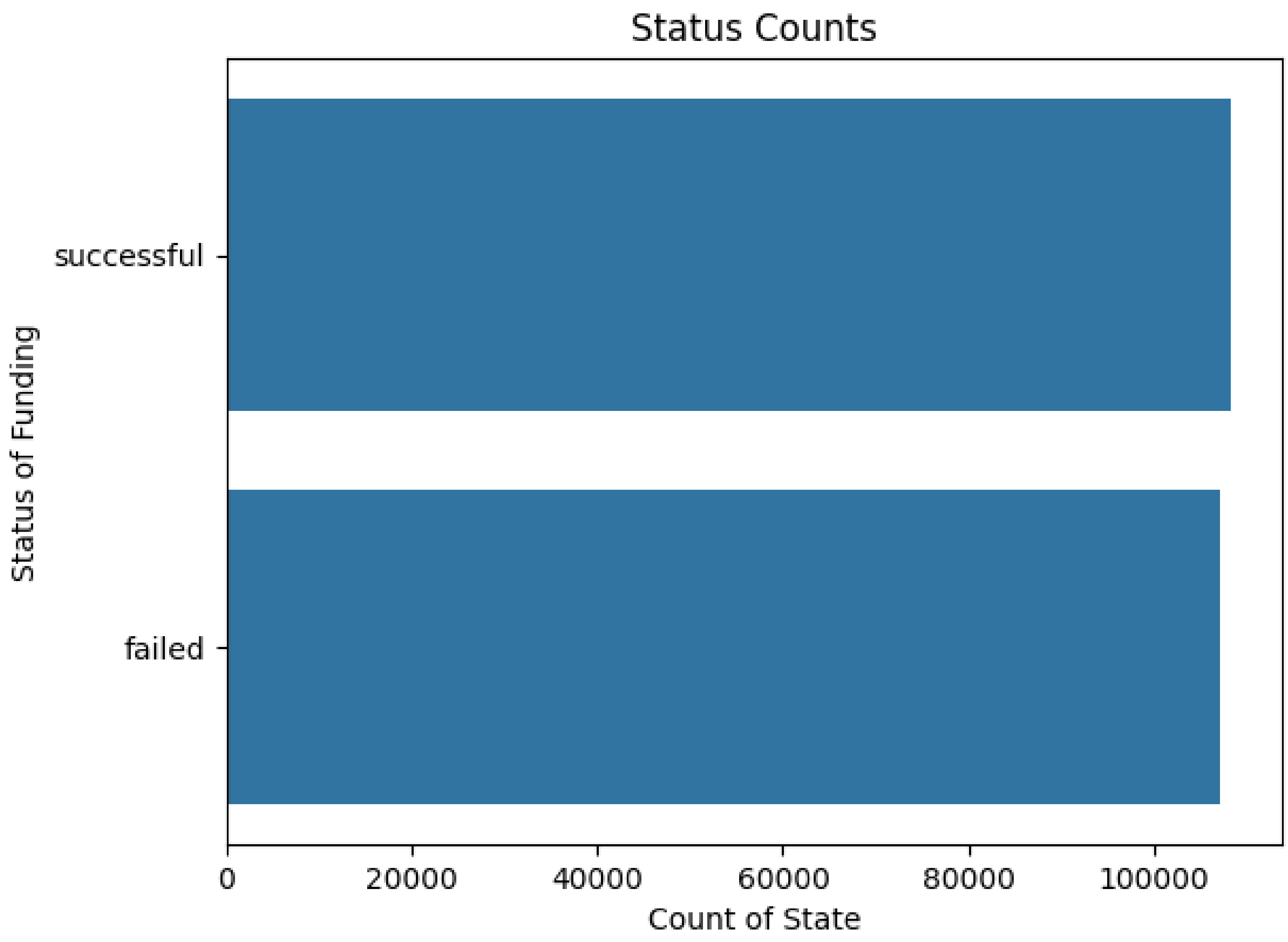
Column name	Type	Description
blurb	Character	Short description of the project
state	Character	Final state of the project ; successful or failed

Exploratory Analysis

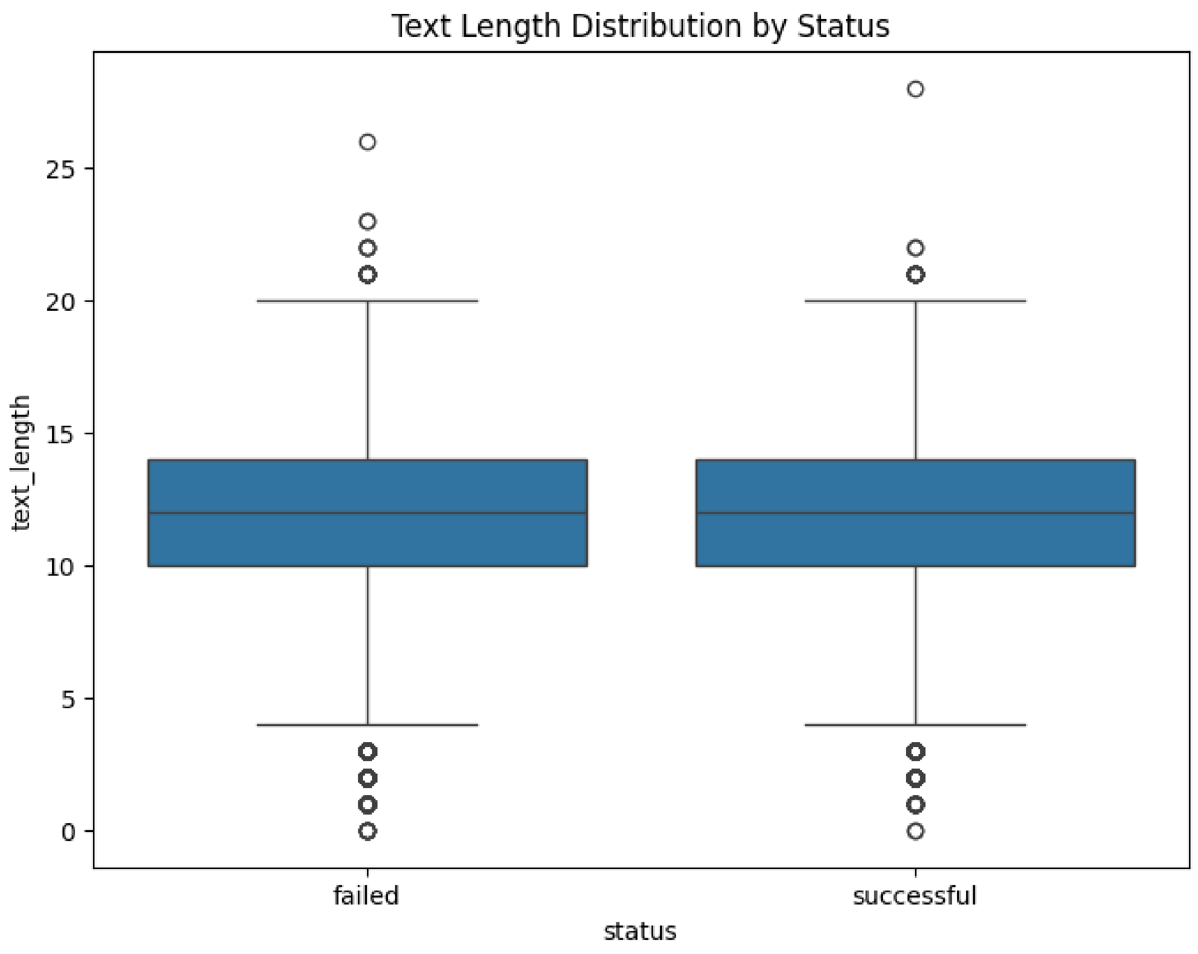
Distribution of Status of Funding.

Both statuses are equally represented in this dataset, so no imbalance.

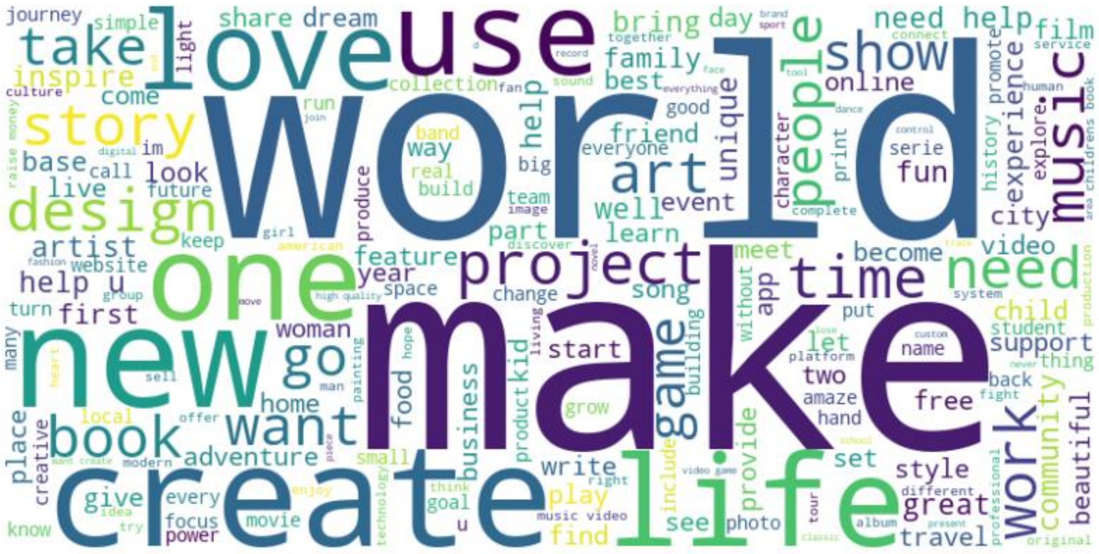
This can provide better accuracy readings in modeling and less bias in training.



Text length of the blurb by status distribution.
About equal lengths of text length found in both classes.
Only a couple outliers.



failed Word Cloud



successful Word Cloud

Failed and successful status word clouds.

	blurb	state	text_clean
0	Using their own character, users go on educati...	failed	Using their own character users go on educatio...
1	MicroFly is a quadcopter packed with WiFi, 6 s...	successful	MicroFly is a quadcopter packed with WiFi 6 se...
2	A small indie press, run as a collective for a...	failed	A small indie press run as a collective for au...
3	Zylor is a new baby cosplayer! Back this kicks...	failed	Zylor is a new baby cosplayer Back this kickst...
4	Hatoful Boyfriend meet Skeletons! A comedy Dat...	failed	Hatoful Boyfriend meet Skeletons A comedy Dati...

Preprocessing Data

First phase of cleaning included removing punctuation.

Continued cleaning text to include tokenization and filtering with stopwords, pronouns, lemmatization to reduce dimensionality.
And finally removing numbers from the clean text .

	blurb	state	text_clean	text_clean_1
0	Using their own character, users go on educati...	failed	Using their own character users go on educatio...	using character users go educational quests ar...
1	MicroFly is a quadcopter packed with WiFi, 6 s...	successful	MicroFly is a quadcopter packed with WiFi 6 se...	microfly quadcopter packed wifi 6 sensors 3 pr...
2	A small indie press, run as a collective for a...	failed	A small indie press run as a collective for au...	small indie press run collective authors want ...
3	Zylor is a new baby cosplayer! Back this kicks...	failed	Zylor is a new baby cosplayer Back this kickst...	zylor new baby cosplayer back kickstarter help...
4	Hatoful Boyfriend meet Skeletons! A comedy Dat...	failed	Hatoful Boyfriend meet Skeletons A comedy Dati...	hatoful boyfriend meet skeletons comedy dating...

	blurb	state	text_clean	text_clean_1	text_clean_2
0	Using their own character, users go on educati...	failed	Using their own character users go on educatio...	using character users go educational quests ar...	use character user go educational quest around...
1	MicroFly is a quadcopter packed with WiFi, 6 s...	successful	MicroFly is a quadcopter packed with WiFi 6 se...	microfly quadcopter packed wifi 6 sensors 3 pr...	microfly quadcopter packed wifi 6 sensor 3 pro...
2	A small indie press, run as a collective for a...	failed	A small indie press run as a collective for au...	small indie press run collective authors want ...	small indie press run collective author want s...
3	Zylor is a new baby cosplayer! Back this kicks...	failed	Zylor is a new baby cosplayer Back this kickst...	zylor new baby cosplayer back kickstarter help...	zylor new baby cosplayer back kickstarter help...
4	Hatoful Boyfriend meet Skeletons! A comedy Dat...	failed	Hatoful Boyfriend meet Skeletons A comedy Dati...	hatoful boyfriend meet skeletons comedy dating...	hatoful boyfriend meet skeleton comedy date si...

	blurb	state	text_clean	text_clean_1	text_clean_2	text_clean_3
0	Using their own character, users go on educati...	failed	Using their own character users go on educatio...	using character users go educational quests ar...	use character user go educational quest around...	use character user go educational quest around...
1	MicroFly is a quadcopter packed with WiFi, 6 s...	successful	MicroFly is a quadcopter packed with WiFi 6 se...	microfly quadcopter packed wifi 6 sensors 3 pr...	microfly quadcopter packed wifi 6 sensor 3 pro...	microfly quadcopter packed wifi sensor proce...
2	A small indie press, run as a collective for a...	failed	A small indie press run as a collective for au...	small indie press run collective authors want ...	small indie press run collective author want s...	small indie press run collective author want s...
3	Zylor is a new baby cosplayer! Back this kicks...	failed	Zylor is a new baby cosplayer Back this kickst...	zylor new baby cosplayer back kickstarter help...	zylor new baby cosplayer back kickstarter help...	zylor new baby cosplayer back kickstarter help...
4	Hatoful Boyfriend meet Skeletons! A comedy Dat...	failed	Hatoful Boyfriend meet Skeletons A comedy Dati...	hatoful boyfriend meet skeletons comedy dating...	hatoful boyfriend meet skeleton comedy date si...	hatoful boyfriend meet skeleton comedy date si...

	text	status
0	use character user go educational quest around...	failed
1	microfly quadcopter packed wifi sensor proce...	successful
2	small indie press run collective author want s...	failed
3	zylor new baby cosplayer back kickstarter help...	failed
4	hatoful boyfriend meet skeleton comedy date si...	failed

The third version of the cleaned text is what we used to set up the new dataframe to go into the next part of the project which is modeling.

```
Topic 0:
world album record story help set new people song friend
Topic 1:
make film time design feature music short video adventure want
Topic 2:
life create work bring like food day start help inspire
Topic 3:
game use help play need base unique love card family
```

	text	status	lda_topic	industry_lda
0	use character user go educational quest around...	failed	3	Games
1	microfly quadcopter packed wifi sensor proce...	successful	0	Music
2	small indie press run collective author want s...	failed	1	Film/Media
3	zylor new baby cosplayer back kickstarter help...	failed	0	Music
4	hatoful boyfriend meet skeleton comedy date si...	failed	0	Music

Model 1
TF-IDF vectorization and LDA topic modeling.

After fitting the LDA model to the tf-idf matrix, we used the LDA topics to assign industry labels to the text-blurbs and mapped the topic indices to industry labels.

```
NMF topic features shape: (215513, 5)
Topic 0:
new album music record song work release city feature way
Topic 1:
make want design use music love im like people unique
Topic 2:
game card play video fun adventure design use base set
Topic 3:
book world story life art create film love series project
Topic 4:
help need album record fund release music bring support raise
```

	text	status	lda_topic	industry_lda	nmf_topic	industry_nmf
0	use character user go educational quest around...	failed	3	Games	3	Publishing/Film/Art
1	microfly quadcopter packed wifi sensor proce...	successful	0	Music	0	Music
2	small indie press run collective author want s...	failed	1	Film/Media	3	Publishing/Film/Art
3	zylor new baby cosplayer back kickstarter help...	failed	0	Music	4	Community
4	hatoful boyfriend meet skeleton comedy date si...	failed	0	Music	3	Publishing/Film/Art

Model 2

TFIDF vectorization and NMF topic modeling.

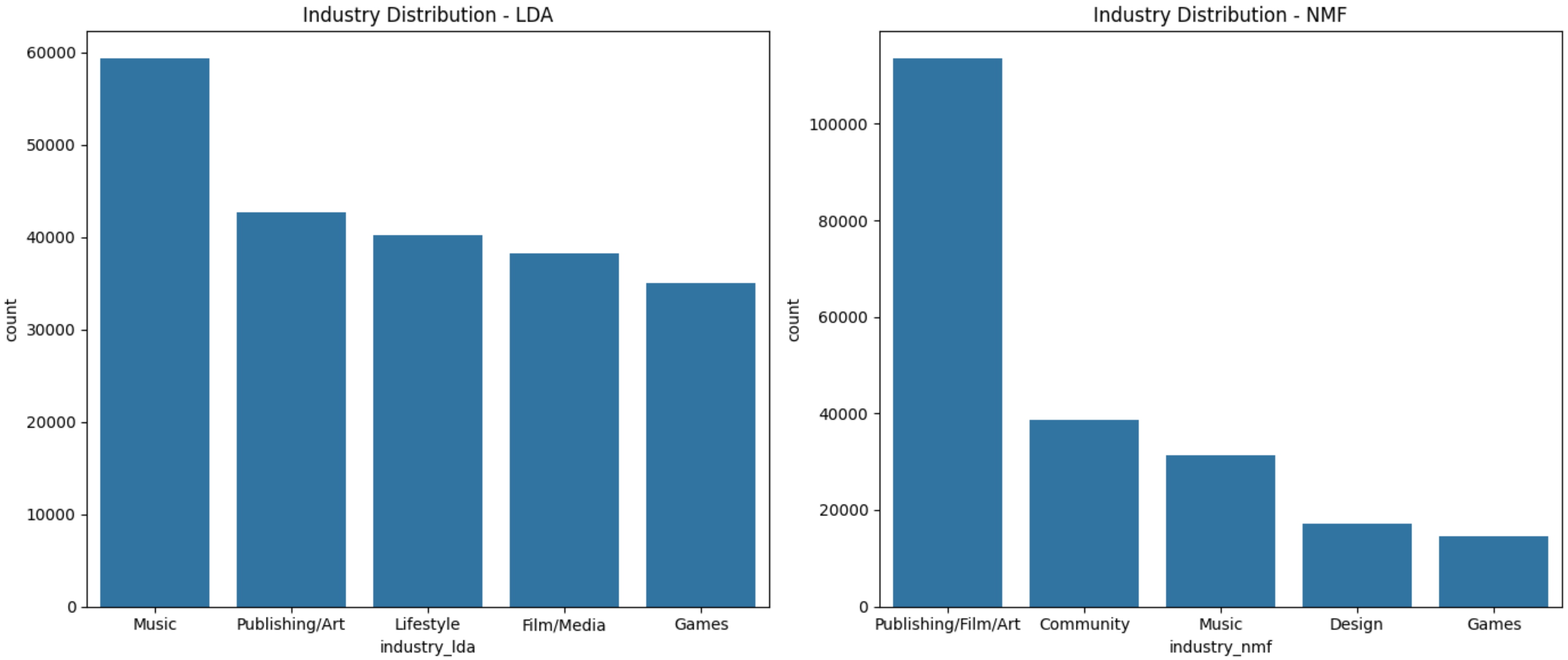
After fitting the NMF model to the tfidf matrix, we used the NMF topics to assign industry labels to the text-blurbs and mapped the topic indices to industry labels.

	text	status	lda_topic	industry_lda	nmf_topic	industry_nmf	industry_combined	industry_combined_id
0	use character user go educational quest around...	failed	3	Games	3	Publishing/Film/Art	Publishing/Film/Art_Games	21
1	microfly quadcopter packed wifi sensor proce...	successful	0	Music	0	Music	Music_Music	18
2	small indie press run collective author want s...	failed	1	Film/Media	3	Publishing/Film/Art	Publishing/Film/Art_Film/Media	20
3	zylor new baby cosplayer back kickstarter help...	failed	0	Music	4	Community	Community_Music	3
4	hatoful boyfriend meet skeleton comedy date si...	failed	0	Music	3	Publishing/Film/Art	Publishing/Film/Art_Music	23

Industry features from both LDA & NMF were combined to from industry_combined_id

Distribution of industry

Overall distribution of industries in both LDA and NMF models. Music, publishing, film, and art were some of the most common industries we saw using this method on our dataset.



```
                                                 text  cluster
0  use character user go educational quest around...        0
1  microfly quadcopter packed wifi  sensor  proce...        2
2  small indie press run collective author want s...        0
3  zylor new baby cosplayer back kickstarter help...        2
4  hatoful boyfriend meet skeleton comedy date si...        0
KMeans feature shape: (215513, 3)
```

KMeansCluster results

MiniBatchKMeans Cluster results

```
                                                 text  mini_cluster
0  use character user go educational quest around...             2
1  microfly quadcopter packed wifi  sensor  proce...             0
2  small indie press run collective author want s...             2
3  zylor new baby cosplayer back kickstarter help...             0
4  hatoful boyfriend meet skeleton comedy date si...             2
Mini KMeans feature shape: (215513, 3)
```

Clustering

Bert embeddings were generated and then applied to the clean_text dataframe. Followed by feeding these embeddings into various models where it was first distributed to KMeans and MiniBatchKMeans for some unsupervised learning.

Supervised Machine Learning Results

Logistic Regression: F1 = 0.668

Best Feature set: Industry_Label_NMF

Random Forest: F1 = 0.634

Best Feature set: BERT_Industry_Label

XGBoost: F1 = 0.649

Best Feature set: bert embeddings

CatBoost: F1 = 0.659

Best Feature set: BERT_Industry_Label

LightGBM: F1 = 0.643

Best Feature set: bert embeddings

	Logistic	RandomForest	XGBoost	LightGBM	CatBoost	Set_Average
BERT_Industry_Label	0.649374	0.634582	0.649236	0.643303	0.659430	0.647185
bert	0.648980	0.634580	0.649585	0.643445	0.658445	0.647007
BERT_Industry_Label_nmf	0.649104	0.634337	0.649324	0.643079	0.659178	0.647004
BERT_Industry_Label_lda	0.649022	0.634428	0.649146	0.643131	0.659287	0.647003
mini_bert	0.649326	0.633613	0.648955	0.642522	0.658777	0.646639
kmeans_bert	0.649341	0.632957	0.648852	0.642882	0.658735	0.646553
kmeans_mini_bert	0.649219	0.631721	0.649527	0.642484	0.659246	0.646439
Industry_Label_LDA	0.563016	0.589102	0.589102	0.589102	0.589102	0.583885
Combined	0.550118	0.592393	0.582919	0.583727	0.565279	0.574887
Vectorized_Industry_Label_comb	0.590911	0.572697	0.571172	0.570664	0.550256	0.571140
Vectorized_Industry_Label_lda	0.565422	0.551308	0.581718	0.589763	0.563058	0.570254
mini	0.552643	0.567416	0.568752	0.569809	0.553808	0.562486
LDA_NMF_Industry_Label	0.532018	0.562945	0.561087	0.571575	0.557748	0.557075
LDA_NMF	0.530905	0.563535	0.562295	0.571735	0.556625	0.557019
kmeans	0.553857	0.559899	0.553307	0.553225	0.548602	0.553778
LDA	0.583516	0.542863	0.536465	0.537990	0.534658	0.547098
LDA_TF_Industry_Labels	0.562338	0.543341	0.534493	0.535840	0.534537	0.542110
NMF	0.412419	0.590193	0.571241	0.578281	0.553858	0.541199
Industry_Labels_Only	0.425329	0.564149	0.565487	0.565487	0.565487	0.537188
NMF_TF_Industry_Labels	0.416674	0.566481	0.553737	0.570303	0.553661	0.532171
Vectorized	0.595868	0.461853	0.495777	0.490938	0.486384	0.506164
Vectorized_Industry_Label_nmf	0.569935	0.466448	0.498098	0.491066	0.499899	0.505089
Industry_Label_NMF	0.668946	0.462518	0.462518	0.462518	0.462518	0.503803

Supervised Machine Learning Results

Logistic Regression: F1 = 0.668

Best Feature set: Industry_Label_NMF

Random Forest: F1 = 0.639

Best Feature set: Industry_Label_LDA

XGBoost: F1 = 0.667

Best Feature set: kmeans_mini_bert

CatBoost: F1 = 0.659

Best Feature set: BERT_Industry_Label

LightGBM: F1 = 0.650

Best Feature set: kmeans_bert

	Logistic	RandomForest	XGBoost	LightGBM	CatBoost	Set_Average
BERT_Industry_Label_nmf	0.649104	0.622542	0.667740	0.650240	0.663208	0.650567
BERT_Industry_Label	0.649374	0.622844	0.667170	0.650447	0.662725	0.650512
BERT_Industry_Label_lda	0.649022	0.622538	0.667237	0.650319	0.662653	0.650354
bert	0.648980	0.623711	0.667012	0.649730	0.661946	0.650276
mini_bert	0.649326	0.621483	0.667057	0.650479	0.661937	0.650056
kmeans_bert	0.649341	0.619385	0.667606	0.650707	0.662290	0.649866
kmeans_mini_bert	0.649219	0.617054	0.667821	0.650503	0.662546	0.649428
Industry_Label_LDA	0.563016	0.639011	0.589102	0.589102	0.589102	0.593866
Vectorized_Industry_Label_lda	0.565422	0.552484	0.580518	0.579737	0.580360	0.571704
Combined	0.550118	0.584012	0.561510	0.577360	0.566144	0.567829
Industry_Label_NMF	0.668946	0.432233	0.462518	0.462518	0.462518	0.497746

Conclusion

- The best performing base model was logistic regression with the NMF topic distributions algorithm applied to industry.